

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278263

Kind Code

A1

Publication Date

September 04, 2025

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System and method for upgrading non-permitted operating systems on standalone computing systems

Abstract

A method includes receiving a request. The request is encrypted to generate an encrypted request. The encrypted request includes a version information for a first software package deployed in a standalone computing system, and a region information and a hardware information for the standalone computing system. The encrypted request is sent to a respective regional data center of a cloud computing system based on the region information of the first software package. A second software package is identified among software packages stored in the respective regional data center. The second software package corresponds to the hardware information of the first software package. The version information of the first software package is compared to a version information of the second software package. In response to identifying that the second software package is more recent than the first software package, the second software package is deployed to the standalone computing system.

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Family ID: 94372828

Appl. No.: 19/200479

Filed: May 06, 2025

Related U.S. Application Data

parent US continuation 18361071 20230728 PENDING child US 19200479

Publication Classification

Int. Cl.: G06F8/65 (20180101); G06F21/57 (20130101)

U.S. Cl.:

CPC G06F8/65 (20130101); G06F21/57 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The application is a continuation of U.S. patent application Ser. No. 18/361,071, filed Jul. 28, 2023, entitled “SYSTEM AND METHOD FOR UPGRADING NON-PERMITTED OPERATING SYSTEMS ON STANDALONE COMPUTING SYSTEMS,” which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to upgrading computing systems, and more specifically to a system and method for upgrading non-permitted operating systems on standalone computing systems.

BACKGROUND

[0003] At times, standalone computing systems of service providers are running on legacy operating systems. It is challenging to upgrade standalone computing systems with latest updates frequently and/or smoothly due to dependency on hardware. Furthermore, it is also challenging to upgrade applications of standalone computing systems along with upgraded operating systems. Standalone computing systems may become vulnerable when operating systems of standalone computing systems reach end-of-life support or non-permitted technology cycles.

SUMMARY

[0004] The system described in the present disclosure provides several practical applications and technical advantages that overcome the current technical problems with upgrading non-permitted operating systems on standalone computing systems.

[0005] In general, a system comprises a cloud computing system operably coupled to a standalone computing system and a user device via a network. The standalone computing system and the cloud computing system are managed by a service provider. In certain embodiments, the service provider may be a financial institution (e.g., a bank). In such embodiments, the standalone computing system may be an automated teller machine (ATM) maintained by a respective bank. Various users are allowed access to the ATM and the cloud computing system. Clients of the bank receive services from the cloud computing system via the ATM. Bank associates manage the ATM and the cloud computing system.

[0006] In operation, the standalone computing system receives a request from a user and encrypts the request to generate an encrypted request. In certain embodiments, in addition to the encrypted information of the request, the encrypted request further comprises a version information for a software package deployed in the standalone computing system, a region information for the standalone computing system, and a hardware information for the standalone computing system. The standalone computing system sends the encrypted request to a respective regional center of the cloud computing system based on the received region information for the standalone computing system. The respective regional center identifies a software package that is stored in the respective regional center, such that the identified software package corresponds to the received hardware information. The respective regional center compares the received version information for the deployed software package to a version information for the identified software package. In

response to determining that the identified software package is more recent than the deployed software package, the respective regional center sends the identified software package to the standalone computing system. The standalone computing system deploys the identified software package. The respective regional center generates an encrypted response based on the encrypted request and sends the encrypted response to the standalone computing system. The standalone computing system decrypts the encrypted response to generate a decrypted response.

[0007] In certain embodiments when the standalone computing system is an ATM, the system may be configured to manage advertisements that are displayed by the ATM. In such embodiments, a user accesses a regional data center of the cloud computing system, where the regional data center corresponds to a desired ATM. A user device of the user displays first advertisements playing on the desired ATM. The user device accesses a virtual application for the desired ATM. The virtual application is configured to emulate the desired ATM. The user device displays second advertisements playing on the virtual application. The user compares the first advertisements to the second advertisements. In response to determining that the first advertisements do not match the second advertisements, the user device identifies a software package corresponding to the desired ATM. The user instructs the user device to generate a software package update based on discrepancy between the first advertisements and the second advertisements. The user device sends the software package update to the regional data center. The regional data center updates the identified software package based on the software package update and sends the updated software package to the desired ATM. The desired ATM deploys the updated software package.

[0008] The system for upgrading non-permitted operating systems on standalone computing systems allows for upgrading standalone computing systems without relying on vendors of the standalone computing system or developers of the non-permitted operating systems by using software packages deployed to the standalone computing systems. The deployed software packages are updated through a cloud computing system that hosts current versions of the software packages. Accordingly, down time of standalone computing systems may be reduced or eliminated. Furthermore, vulnerabilities due to using non-permitted operating systems may be reduced or eliminated, which in turn improves security of standalone computing systems. In certain embodiments when the standalone computing system is an ATM, the system also allows for managing advertisements that are displayed by the ATM.

[0009] Accordingly, the following disclosure is particularly integrated into practical applications of: (1) upgrading non-permitted operating systems on standalone computing systems through a cloud computing system; (2) reducing or eliminating down time of standalone computing systems; and (3) improving security of standalone computing systems.

[0010] In one embodiment, a system includes a standalone computing system and a cloud computing system communicatively coupled to the standalone computing system. The standalone computing system includes a first memory and a first processor communicatively coupled to the first memory. The first memory is configured to store a first software package. The first software package includes a first hardware management module and a first client application. The first processor is configured to receive a request from a user and encrypt the request to generate an encrypted request. The encrypted request includes a version information for the first software package, a region information for the standalone computing system, and a hardware information for the standalone computing system. The first processor is further configured to send the encrypted request to a respective regional data center of a cloud computing system based on the region information of the first software package. In response to sending the encrypted request, the first processor is further configured to receive a second software package from the respective regional data center and deploy the second software package. The cloud computing system includes a plurality of regional data centers. The respective regional data center includes a second memory and a second processor communicatively coupled to the second memory. The second memory is configured to store a plurality of software packages. Each software package corresponds to a

respective hardware configuration of the standalone computing system. Each software package includes a hardware management module and a client application. The second processor is configured to receive the encrypted request from the standalone computing system and identify the second software package among the plurality of software packages. The second software package corresponds to the hardware information of the first software package. The second processor is further configured to compare the version information of the first software package to a version information of the second software package. In response to identifying that the second software package is more recent than the first software package, The second processor is further configured to send the second software package to the standalone computing system.

[0011] Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, where like reference numerals represent like parts.

[0013] FIG. 1 illustrates an embodiment of a system for upgrading non-permitted operating systems on standalone computing systems;

[0014] FIG. 2 illustrates an example operational flow of the system of FIG. 1 for upgrading non-permitted operating systems on standalone computing systems; and

[0015] FIG. 3 illustrates an example operational flow of the system of FIG. 1 for managing advertisements displayed by an automated teller machine (ATM).

DETAILED DESCRIPTION

[0016] As described above, previous technologies fail to provide effective systems and methods for upgrading non-permitted operating systems on standalone computing systems. Embodiments of the present disclosure and their advantages may be understood by referring to FIGS. 1, 2 and 3. FIGS. 1, 2 and 3 are used to describe a system and method for upgrading non-permitted operating systems on standalone computing systems.

System Overview

[0017] FIG. 1 illustrates an embodiment of a system **100** for upgrading non-permitted operating systems on standalone computing systems. In certain embodiments, the system **100** comprises a cloud computing system **134** operably coupled to a standalone computing system **106** and a user device **168** via a network **102**. Network **102** enables the communication between the components of the system **100**. The cloud computing system **134** provides services to a user **104a** via the standalone computing system **106**. A user **104b** communicates with the cloud computing system **134** via the user device **168**. The standalone computing system **106** and the cloud computing system **134** are managed by a service provider. In certain embodiments, the service provider may be a financial institution (e.g., a bank). In such embodiments, the standalone computing system **106** may be an automated teller machine (ATM), the user **104a** may be a client of a bank, which receives services from the cloud computing system **134** via the ATM, and the user **104b** may be a bank associate, which manages the ATM and the cloud computing system **134**. In other embodiments, the system **100** may not have all the components listed and/or may have other elements instead of, or in addition to, those listed above.

[0018] In general, the standalone computing system **106** receives a request **128** from the user **104a** and encrypts the request **128** to generate an encrypted request **164**. In certain embodiments, in addition to the encrypted information of the request **128**, the encrypted request **164** further

comprises a version information **122** for a software package **116** deployed in the standalone computing system **106**, a region information **124** for the standalone computing system **106**, and a hardware information **126** for the standalone computing system **106**. The standalone computing system **106** sends the encrypted request **164** to a respective regional center (e.g., regional data center **136a**) of the cloud computing system **134** based on the received region information **124**. The respective regional center (e.g., regional data center **136a**) identifies a software package (e.g., software package **152a**) that is stored in the respective regional center (e.g., regional data center **136a**) such that the identified software package (e.g., software package **152a**) corresponds to the received hardware information **126**. The respective regional center (e.g., regional data center **136a**) compares the received version information **122** for the deployed software package **116** to a version information (e.g., version information **158a**) for the identified software package (e.g., software package **152a**). In response to determining that the identified software package (e.g., software package **152a**) is more recent than the deployed software package **116**, the respective regional center (e.g., regional data center **136a**) sends the identified software package (e.g., software package **152a**) to the standalone computing system **106**. The standalone computing system **106** deploys the identified software package (e.g., software package **152a**). The respective regional center (e.g., regional data center **136a**) generates an encrypted response **166** based on the encrypted request **164** and sends the encrypted response **166** to the standalone computing system **106**. The standalone computing system **106** decrypts the encrypted response **166** to generate a decrypted response **130**.

[0019] In certain embodiments when the standalone computing system **106** is an ATM, the system **100** may be configured to manage advertisements that are displayed by the ATM. In such embodiments, the user device **168** of the user **104b** accesses a regional data center (e.g., regional data center **136b**) of the cloud computing system **134**, where the regional data center (e.g., a regional data center **136b**) corresponds to a desired ATM (e.g., standalone computing system **106**). The user device **168** displays first advertisements **186** playing on the desired ATM (e.g., standalone computing system **106**). The user device **168** accesses a virtual application (e.g., VM application **184**) for the desired ATM (e.g., standalone computing system **106**). The virtual application (e.g., VM application **184**) is configured to emulate the desired ATM (e.g., standalone computing system **106**). The user device **168** displays second advertisements **188** playing on the virtual application (e.g., VM application **184**). The user **104b** compares the first advertisements **186** to the second advertisements **188**. In response to determining that the first advertisements **186** do not match the second advertisements **188**, the user device **168** identifies a software package (e.g., software package **152b**) corresponding to the desired ATM (e.g., standalone computing system **106**). The user **104b** instructs the user device **168** to generate a software package update **190** based on discrepancy between the first advertisements **186** and the second advertisements **188**. The user device **168** sends the software package update **190** to the regional data center (e.g., regional data center **136b**). The regional data center (e.g., regional data center **136b**) updates the identified software package (e.g., software package **152b**) based on the software package update **190** and sends the updated software package **132** to the desired ATM (e.g., standalone computing system **106**). The desired ATM (e.g., standalone computing system **106**) deploys the updated software package **132**.

System Components

Network

[0020] Network **102** may be any suitable type of wireless and/or wired network. The network **102** may or may not be connected to the Internet or public network. The network **102** may include all or a portion of an Intranet, a peer-to-peer network, a switched telephone network, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), a wireless PAN (WPAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a

plain old telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, etc.), a long-term evolution (LTE) network, a universal mobile telecommunications system (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a near field communication (NFC) network, and/or any other suitable network. The network **102** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

Standalone Computing System

[0021] The standalone computing system **106** is generally any device that is configured to process data and communicate with other components of the system **100** via the network **102**. The standalone computing system **106** may comprise a processor **108** in signal communication with a memory **114** and a network interface **110**. The standalone computing system **106** may include a user interface, such as a display **112**, a microphone, keypad, or other appropriate terminal equipment usable by the user **104a**. In certain embodiments, the standalone computing system **106** is an ATM.

[0022] Processor **108** comprises one or more processors operably coupled to the memory **114**. Processor **108** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). Processor **108** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, processor **108** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions (e.g., software package **116**) and perform one or more functions described herein.

[0023] Network interface **110** is configured to enable wired and/or wireless communications (e.g., via network **102**). Network interface **110** is configured to communicate data between the standalone computing system **106** and other components of the system **100**. For example, the network interface **110** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. Processor **108** is configured to send and receive data using the network interface **110**. Network interface **110** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

[0024] Memory **114** comprises a non-transitory computer-readable medium such as one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory **114** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **114** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **114** may store any of the information described in FIGS. **1**, **2** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein. Memory **114** is operable to store software instructions (e.g., software package **116**), and/or any other data and instructions. The software instructions may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **108**.

[0025] In certain embodiments, the software package **116** comprises a hardware management module **118** and a client application **120**. The hardware management module **118**, when executed by the processor **108**, allows the processor **108** to control the hardware of the standalone computing system **106**. The client application **120**, when executed by the processor **108**, allows the processor **108** to communicate with the user **104a** and the cloud computing system **134**. The software

package **116** may further comprise a version information **122** of the software package **116**, a region information **124** for the standalone computing system **106**, and a hardware information **126** for the standalone computing system **106**. The version information **122** of the software package **116** may comprise a version number, such that a greater version number corresponds to a more recent version of the software package **116**. The region information **124** for the standalone computing system **106** may include any information identifying a location of the standalone computing system **106**, for example, an IP address of the standalone computing system **106**. The hardware information **126** for the standalone computing system **106** may include a processor clock speed, a number of processor cores, a screen resolution, a total memory, or the like.

[0026] In operation, the processor **108** of the standalone computing system **106** is configured to receive a request **128** from a user **104a**. In certain embodiments when the standalone computing system **106** is an ATM, the request **128** may comprise a request to perform a desired interaction with the ATM. The processor **108** of the standalone computing system **106** encrypts the request **128** to generate an encrypted request **164**. In certain embodiments, the processor **108** of the standalone computing system **106** may execute any appropriate encryption algorithm to perform the encryption. In certain embodiments, in addition to the encrypted information of the request **128**, the encrypted request **164** further comprises a version information **122** for a software package **116** that is deployed in the standalone computing system **106**, a region information **124** for the standalone computing system **106**, and a hardware information **126** for the standalone computing system **106**. The version information **122** of the software package **116** may comprise a version number, such that a greater version number corresponds to a more recent version of the deployed software package **116**. The region information **124** for the standalone computing system **106** may include any information identifying a location of the standalone computing system **106**, for example, an IP address of the standalone computing system **106**. The hardware information **126** for the standalone computing system **106** may include a processor clock speed, a number of processor cores, a screen resolution, a total memory, or the like.

[0027] The processor **108** of the standalone computing system **106** sends the encrypted request **164** to a respective regional center (e.g., regional data center **136a**) of the cloud computing system **134** based on the received region information **124**. The processor **108** of the standalone computing system **106** receives a software package (e.g., software package **152a** of FIG. 1) from the respective regional center (e.g., regional data center **136a**), such that software package (e.g., software package **152a** of FIG. 1) is more recent than the deployed software package **116**. The processor **108** of the standalone computing system **106** deploys the received software package (e.g., software package **152a**). In certain embodiments, the deployment process comprises replacing the deployed software package **116** with the received software package (e.g., software package **152a**). The processor **108** of the standalone computing system **106** receives an encrypted response **166** from the respective regional center (e.g., regional data center **136a**) and decrypts the encrypted response **166** to generate a decrypted response **130**. In certain embodiments, the processor **108** of the standalone computing system **106** may execute any appropriate decryption algorithm to perform the decryption.

[0028] In certain embodiments when the standalone computing system **106** is an ATM, the ATM (e.g., standalone computing system **106**) receives the updated software package **132** from a regional data center (e.g., regional data center **136b**) of the cloud computing system **134**. A processor (e.g., processor **108**) of the ATM (e.g., standalone computing system **106**) deploys the updated software package **132**. In certain embodiments, the deployment process comprises replacing a software package **116** deployed in the ATM (e.g., standalone computing system **106**) with the updated software package **132**.

Cloud Computing System

[0029] The cloud computing system **134** is configured to provide various services to the user **104a** via the standalone computing system **106**. The cloud computing system **134** may include a plurality

of regional data centers, such as regional data centers **136a** and **136b**. In the illustrated embodiment, the cloud computing system **134** comprises two regional data centers. In other embodiments, the cloud computing system **134** may comprise more than two regional data centers. [0030] The regional data center **136a** (**136b**) is generally any device that is configured to process data and communicate with other components of the system **100** via the network **102**. The regional data center **136a** (**136b**) may comprise a processor **138a** (**138b**) in signal communication with a memory **142a** (**142b**) and a network interface **140a** (**140b**).

[0031] Processor **138a** (**138b**) comprises one or more processors operably coupled to the memory **142a** (**142b**). Processor **138a** (**138b**) is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). Processor **138a** (**138b**) may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, processor **138a** (**138b**) may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions **144a** (**144b**) and an administrator application **146a** (**146b**), and perform one or more functions described herein.

[0032] Network interface **140a** (**140b**) is configured to enable wired and/or wireless communications (e.g., via network **102**). Network interface **140a** (**140b**) is configured to communicate data between the regional data center **136a** (**136b**) and other components of the system **100**. For example, the network interface **140a** (**140b**) may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. Processor **138a** (**138b**) is configured to send and receive data using the network interface **140a** (**140b**). Network interface **140a** (**140b**) may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

[0033] Memory **142a** (**142b**) comprises a non-transitory computer-readable medium such as one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory **142a** (**142b**) may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **142a** (**142b**) may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **142a** (**142b**) may store any of the information described in FIGS. **1**, **2** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein. Memory **142a** (**142b**) is operable to store software instructions **144a** (**144b**) and an administrator application **146a** (**146b**), and/or any other data and instructions. The software instructions **144a** (**144b**) and the administrator application **146a** (**146b**) may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **138a** (**138b**).

[0034] Memory **142a** (**142b**) is further operable to store a regional database **148a** (**148b**), which comprises regional data **150a** (**150b**). In certain embodiments when the standalone computing system **106** is configured access the regional data center **136a**, the standalone computing system **106** accesses the regional data **150a** of the regional database **148a**. In certain embodiments when the standalone computing system **106** is configured access the regional data center **136b**, the standalone computing system **106** accesses the regional data **150b** of the regional database **148b**.

[0035] Memory **142a** (**142b**) is further operable to store software packages **152a** (**152b**). The software packages **152a** (**152b**) are most recent versions of software packages. Each software package **152a** (**152b**) comprises a hardware management module **154a** (**154b**) and a client

application **156a (156b)**. The software package **152a (152b)** may further comprise a version information **158a (158b)** of the software package **152a (152b)**, a region information **160a (160b)** for a respective standalone computing system (e.g., standalone computing system **106**), and a hardware information **162a (162b)** for the respective standalone computing system. The version information **158a (158b)** of the software package **152a (152b)** may comprise a version number, such that a greater version number corresponds to a more recent version of the software package **152a (152b)**. The region information **160a (160b)** may include any information identifying a location of the respective standalone computing system, for example, an IP address. The hardware information **162a (162b)** for the respective standalone computing system may include a processor clock speed, a number of processor cores, a screen resolution, a total memory, or the like.

[0036] In operation, a processor (e.g., processor **138a**) of a regional center (e.g., regional data center **136a**) of the cloud computing system **134** receives the encrypted request **164** from the standalone computing system **106**. The processor (e.g., processor **138a**) of the respective regional center (e.g., regional data center **136a**) identifies a software package (e.g., software package **152a**) that is stored in the respective regional center (e.g., regional data center **136a**) such that the identified software package (e.g., software package **152a**) corresponds to the received hardware information **126**. In certain embodiments, the processor (e.g., processor **138a**) of the respective regional center (e.g., regional data center **136a**) identifies the identified software package (e.g., software package **152a** of) by determining that the received hardware information **126** matches a hardware information (e.g., hardware information **162a**) of the identified software package (e.g., software package **152a**). The processor (e.g., processor **138a**) of the respective regional center (e.g., regional data center **136a**) compares the received version information **122** for the deployed software package **116** to a version information (e.g., version information **158a**) for the identified software package (e.g., software package **152a**). In response to determining that the identified software package (e.g., software package **152a**) is more recent than the deployed software package **116**, the processor (e.g., processor **138a**) of the respective regional center (e.g., regional data center **136a**) sends the identified software package (e.g., software package **152a**) to the standalone computing system **106**. In certain embodiments, the processor (e.g., processor **138a**) of the respective regional center (e.g., regional data center **136a**) encrypts the identified software package (e.g., software package **152a** of FIG. 1) before sending it to the standalone computing system **106**. [0037] In certain embodiments, a processor (e.g., processor **138b**) of a regional data center (e.g., regional data center **136b**) receives the software package update **190** from the user device **168**, updates a software package (e.g., software package **152b**) based on the software package update **190**, and sends the updated software package **132** to the standalone computing system **106**. In certain embodiments, the processor (e.g., processor **138b**) of the regional center (e.g., regional data center **136b**) encrypts the updated software package **132** before sending it to the standalone computing system **106**.

User Device

[0038] The user device **168** is generally any device that is configured to process data and communicate with other components of the system **100** via the network **102**. The user **104b** communicates with the cloud computing system **134** via the user device **168**. The user device **168** may comprise a processor **170** in signal communication with a memory **180** and a network interface **172**. The user device **168** may include a user interface, such as a display **174**, a microphone, keypad, or other appropriate terminal equipment usable by the user **104b**.

[0039] Processor **170** comprises one or more processors operably coupled to the memory **180**. Processor **170** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). Processor **170** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are

configured to process data and may be implemented in hardware or software. For example, processor **170** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions **182** and a virtual machine (VM) application **184**, and perform one or more functions described herein.

[0040] Network interface **172** is configured to enable wired and/or wireless communications (e.g., via network **102**). Network interface **172** is configured to communicate data between the user device **168** and other components of the system **100**. For example, the network interface **172** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. Processor **170** is configured to send and receive data using the network interface **172**. Network interface **172** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

[0041] Memory **180** comprises a non-transitory computer-readable medium such as one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory **180** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **180** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **180** may store any of the information described in FIGS. **1**, **2** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein. The memory **180** is operable to store software instructions **182**, the virtual machine (VM) application **184**, and/or any other data and instructions. The software instructions **182** and the virtual machine (VM) application **184** may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **170**.

[0042] In operation, the user **104b** accesses a regional data center (e.g., regional data center **136b**) of the cloud computing system **134**, where the regional data center (e.g., a regional data center **136b**) corresponds to a desired ATM (e.g., standalone computing system **106**). In certain embodiments, a user device **168** of the user **104b** accesses the regional data center (e.g., regional data center **136b**) via an administrator application (e.g., administrator application **146b**) of the regional data center (e.g., regional data center **136b** of FIG. **1**). The processor **170** of the user device **168** displays first advertisements **186** playing on the desired ATM (e.g., standalone computing system **106**). In certain embodiments, the processor **170** of the user device **168** instructs a display **174** of the user device **168** to display the first advertisements **186** in a first window **176**. The processor **170** of the user device **168** accesses a virtual application (e.g., VM application **184**) for the desired ATM (e.g., standalone computing system **106**). The virtual application (e.g., VM application **184**) is configured to emulate the desired ATM (e.g., standalone computing system **106**). The processor **170** of the user device **168** displays second advertisements **188** playing on the virtual application (e.g., VM application **184**). In certain embodiments, the processor **170** of the user device **168** instructs the display **174** of the user device **168** to display the second advertisements **188** in a second window **178**. The user **104b** compares the first advertisements **186** to the second advertisements **188**.

[0043] In response to determining that the first advertisements **186** do not match the second advertisements **188**, the processor **170** of the user device **168** identifies a software package (e.g., software package **152b**) corresponding to the desired ATM (e.g., standalone computing system **106**). In certain embodiments, the processor **170** of the user device **168** identifies the software package (e.g., software package **152b**) by comparing a hardware information **126** for a deployed software package **116** of the desired ATM (e.g., standalone computing system **106**) and a hardware information **162b** for the identified software package (e.g., software package **152b**).

[0044] The user **104b** instructs the processor **170** of the user device **168** to generate a software package update **190** based on discrepancy between the first advertisements **186** and the second advertisements **188**. In certain embodiments when content of the first advertisements **186** is different from content of the second advertisements **188**, the software package update **190** comprises updates to a client application (e.g., client application **156b**) of the identified software package (e.g., software package **152b**). In certain embodiments when the content of the first advertisements **186** is same as the content of the second advertisements **188** but they are displayed in different manner due to screen resolution mismatch, the software package update **190** comprises updates to a hardware management module (e.g., hardware management module **154b**) of the identified software package (e.g., software package **152b**). The processor **170** of the user device **168** sends the software package update **190** to the regional data center (e.g., regional data center **136b**).

Example Method for Upgrading Non-Permitted Operating Systems on Standalone Computing Systems

[0045] FIG. 2 illustrates an example flowchart of a method **200** for upgrading non-permitted operating systems on standalone computing systems. Modifications, additions, or omissions may be made to method **200**. Method **200** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. For example, one or more operations of method **200** may be implemented, at least in part, in the form of the software instructions (e.g., software package **116**, instructions **144a**, **144b**, and **182**, administrator applications **146a** and **146b**, and/or VM application **184** of FIG. 1), stored on non-transitory, tangible, computer-readable medium (e.g., memories **114**, **142a**, **142b**, and/or **180** of FIG. 1) that when executed by one or more processors (e.g., processors **108**, **138a**, **138b**, and/or **170** of FIG. 1) may cause the one or more processors to perform operations **202-222**.

[0046] Method **200** starts with operation **202**, where a processor **108** of a standalone computing system **106** receives a request **128** from a user **104a**. In certain embodiments when the standalone computing system **106** is an ATM, the request **128** may comprise a request to perform a desired interaction with the ATM.

[0047] At operation **204**, the processor **108** of the standalone computing system **106** encrypts the request **128** to generate an encrypted request **164**. In certain embodiments, the processor **108** of the standalone computing system **106** may execute any appropriate encryption algorithm to perform encryption. In certain embodiments, in addition to the encrypted information of the request **128**, the encrypted request **164** further comprises a version information **122** for a deployed software package **116**, a region information **124** for the standalone computing system **106**, and a hardware information **126** for the standalone computing system **106**. The version information **122** of the software package **116** may comprise a version number, such that a greater version number corresponds to a more recent version of the deployed software package **116**. The region information **124** for the standalone computing system **106** may include any information identifying a location of the standalone computing system **106**, for example, an IP address of the standalone computing system **106**. The hardware information **126** for the standalone computing system **106** may include a processor clock speed, a number of processor cores, a screen resolution, a total memory, or the like.

[0048] At operation **206**, the processor **108** of the standalone computing system **106** sends the encrypted request **164** to a respective regional center (e.g., regional data center **136a** of FIG. 1) of a cloud computing system **134** based on the received region information **124**.

[0049] At operation **208**, a processor (e.g., processor **138a** of FIG. 1) of the respective regional center (e.g., regional data center **136a** of FIG. 1) identifies a software package (e.g., software package **152a** of FIG. 1) that is stored in the respective regional center (e.g., regional data center **136a** of FIG. 1). The identified software package (e.g., software package **152a** of FIG. 1) corresponds to the received hardware information **126**. In certain embodiments, the processor (e.g.,

processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) identifies the identified software package (e.g., software package **152a** of FIG. **1**) by determining that the received hardware information **126** matches a hardware information (e.g., hardware information **162a** of FIG. **1**) of the identified software package (e.g., software package **152a** of FIG. **1**).

[0050] At operation **210**, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) compares the received version information **122** for the deployed software package **116** to a version information (e.g., version information **158a** of FIG. **1**) for the identified software package (e.g., software package **152a** of FIG. **1**).

[0051] At operation **212**, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) determines if the identified software package (e.g., software package **152a** of FIG. **1**) is more recent than the deployed software package **116**.

[0052] In response to determining at operation **212** that the identified software package (e.g., software package **152a** of FIG. **1**) is not more recent than the deployed software package **116**, method **200** continues to operation **218**.

[0053] In response to determining at operation **212** that the identified software package (e.g., software package **152a** of FIG. **1**) is more recent than the deployed software package **116**, method **200** continues to operation **214**. At operation **214**, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) sends the identified software package (e.g., software package **152a** of FIG. **1**) to the standalone computing system **106**. In certain embodiments, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) encrypts the identified software package (e.g., software package **152a** of FIG. **1**) before sending it to the standalone computing system **106**.

[0054] At operation **216**, the processor **108** of the standalone computing system **106** deploys the identified software package (e.g., software package **152a** of FIG. **1**). In certain embodiments, the deployment process comprises replacing the deployed software package **116** with the identified software package (e.g., software package **152a** of FIG. **1**).

[0055] After performing operation **216** or in response to determining at operation **212** that the identified software package (e.g., software package **152a** of FIG. **1**) is not more recent than the deployed software package **116**, method **200** continues to operation **218**. At operation **218**, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) generates an encrypted response **166** based on the encrypted request **164**.

[0056] At operation **220**, the processor (e.g., processor **138a** of FIG. **1**) of the respective regional center (e.g., regional data center **136a** of FIG. **1**) sends the encrypted response **166** to the standalone computing system **106**.

[0057] At operation **222**, the processor **108** of the standalone computing system **106** decrypts the encrypted response **166** to generate a decrypted response **130**. In certain embodiments, the processor **108** of the standalone computing system **106** may execute any appropriate decryption algorithm to perform decryption. After performing operation **222**, method **200** ends.

Example Method for Managing Advertisements Displayed by an Automated Teller Machine (ATM)

[0058] FIG. **3** illustrates an example flowchart of a method **300** for managing advertisements displayed by an automated teller machine (ATM). Modifications, additions, or omissions may be made to method **300**. Method **300** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. For example, one or more operations of method **300** may be implemented, at least in part, in the form of the software instructions (e.g., software package **116**, instructions **144a**, **144b**, and **182**, administrator applications **146a** and **146b**, and/or VM application **184** of FIG. **1**), stored on non-transitory, tangible, computer-readable medium (e.g., memories **114**, **142a**, **142b**, and/or **180** of FIG. **1**) that when executed by one or more processors (e.g., processors **108**, **138a**, **138b**, and/or **170** of FIG. **1**) may cause the one or more processors to perform operations **302-324**.

[0059] Method **300** starts with operation **302**, where a user **104b** accesses a regional data center (e.g., regional data center **136b** of FIG. **1**) of a cloud computing system **134**, where the regional data center (e.g., a regional data center **136b** of FIG. **1**) corresponds to a desired ATM (e.g., standalone computing system **106** of FIG. **1**). In certain embodiments, a user device **168** of the user **104b** accesses the regional data center (e.g., regional data center **136b** of FIG. **1**) via an administrator application (e.g., administrator application **146b** of FIG. **1**) of the regional data center (e.g., regional data center **136b** of FIG. **1**).

[0060] At operation **304**, a processor **170** of the user device **168** displays first advertisements **186** playing on the desired ATM (e.g., standalone computing system **106** of FIG. **1**). In certain embodiments, the processor **170** of the user device **168** instructs a display **174** of the user device **168** to display the first advertisements **186** in a first window **176**.

[0061] At operation **306**, the processor **170** of the user device **168** accesses a virtual application (e.g., VM application **184** of FIG. **1**) for the desired ATM (e.g., standalone computing system **106** of FIG. **1**). The virtual application (e.g., VM application **184** of FIG. **1**) is configured to emulate the desired ATM (e.g., standalone computing system **106** of FIG. **1**).

[0062] At operation **308**, the processor **170** of the user device **168** displays second advertisements **188** playing on the virtual application (e.g., VM application **184** of FIG. **1**). In certain embodiments, the processor **170** of the user device **168** instructs the display **174** of the user device **168** to display the second advertisements **188** in a second window **178**.

[0063] At operation **310**, the user **104b** compares the first advertisements **186** to the second advertisements **188**.

[0064] At operation **312**, the user **104b** determines if the first advertisements **186** match the second advertisements **188**. In response to determining at operation **312** that the first advertisements **186** match the second advertisements **188**, method **300** ends.

[0065] In response to determining at operation **312** that the first advertisements **186** do not match the second advertisements **188**, method **300** continues to operation **314**. At operation **314**, the processor **170** of the user device **168** identifies a software package (e.g., software package **152b** of FIG. **1**) corresponding to the desired ATM (e.g., standalone computing system **106** of FIG. **1**). In certain embodiments, the processor **170** of the user device **168** identifies the software package (e.g., software package **152b** of FIG. **1**) by comparing a hardware information **126** for a deployed software package **116** of the desired ATM (e.g., standalone computing system **106** of FIG. **1**) and a hardware information **162b** for the identified software package (e.g., software package **152b** of FIG. **1**).

[0066] At operation **316**, the user **104b** instructs the processor **170** of the user device **168** to generate a software package update **190** based on discrepancy between the first advertisements **186** and the second advertisements **188**. In certain embodiments when content of the first advertisements **186** is different from content of the second advertisements **188**, the software package update **190** comprises updates to a client application (e.g., client application **156b** of FIG. **1**) of the identified software package (e.g., software package **152b** of FIG. **1**). In certain embodiments when the content of the first advertisements **186** is same as the content of the second advertisements **188** but they are displayed in different manner due to screen resolution mismatch, the software package update **190** comprises updates to a hardware management module (e.g., hardware management module **154b** of FIG. **1**) of the identified software package (e.g., software package **152b** of FIG. **1**).

[0067] At operation **318**, the processor **170** of the user device **168** sends the software package update **190** to the regional data center (e.g., regional data center **136b** of FIG. **1**).

[0068] At operation **320**, a processor (e.g., processor **138b** of FIG. **1**) of the regional data center (e.g., regional data center **136b** of FIG. **1**) updates the identified software package (e.g., software package **152b** of FIG. **1**) based on the software package update **190**.

[0069] At operation **322**, the processor (e.g., processor **138b** of FIG. **1**) of the regional data center

(e.g., regional data center **136b** of FIG. **1**) sends the updated software package **132** to the desired ATM (e.g., standalone computing system **106** of FIG. **1**). In certain embodiments, the processor (e.g., processor **138b** of FIG. **1**) of the regional center (e.g., regional data center **136b** of FIG. **1**) encrypts the updated software package **132** before sending it to the desired ATM (e.g., standalone computing system **106** of FIG. **1**).

[0070] At operation **324**, a processor (e.g., processor **108** of FIG. **1**) of the desired ATM (e.g., standalone computing system **106** of FIG. **1**) deploys the updated software package **132**. In certain embodiments, the deployment process comprises replacing the deployed software package **116** with the updated software package **132**. After performing operation **324**, method **300** ends.

[0071] While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

[0072] In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

[0073] To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a standalone computing system comprising: a first memory configured to store a first software package; and a first processor communicatively coupled to the first memory, wherein the first processor is configured to: receive a request from a user; encrypt the request to generate an encrypted request, wherein the encrypted request comprises a version information for the first software package, a region information for the standalone computing system, and a hardware information for the standalone computing system; send the encrypted request to a respective regional data center of a cloud computing system based on the region information of the first software package; in response to sending the encrypted request, receive a second software package from the respective regional data center; and deploy the second software package; and the cloud computing system communicatively coupled to the standalone computing system, the cloud computing system comprising a plurality of regional data centers, wherein the respective regional data center comprises: a second memory configured to store a plurality of software packages, wherein each software package corresponds to a respective hardware configuration of the standalone computing system; and a second processor communicatively coupled to the second memory, wherein the second processor is configured to: receive the encrypted request from the standalone computing system; identify the second software package among the plurality of software packages, wherein the second software package corresponds to the hardware information of the first software package; compare the version information of the first software package to a version information of the second software package; and in response to identifying that the second software package is more recent than the first software package, send the second software package

to the standalone computing system.

2. The system of claim 1, wherein: the second processor is further configured to: generate an encrypted response based on the encrypted request; and send the encrypted response to the standalone computing system; and the first processor is further configured to: receive the encrypted response from the respective regional data center; and decrypt the encrypted response.
3. The system of claim 1, wherein the first hardware management module of the first software package is different from a second hardware management module of the second software package.
4. The system of claim 1, wherein the first software package is different from the second software package.
5. The system of claim 1, wherein the standalone computing system comprises an automated teller machine (ATM).
6. The system of claim 1, wherein deploying the second software package comprises replacing the first software package with the second software package.
7. The system of claim 1, wherein the second processor is configured to: before sending the second software package to the standalone computing system, encrypt the second software package.
8. A method comprising: receiving a request from a user; encrypting the request to generate an encrypted request, wherein the encrypted request comprises: a version information for a first software package deployed in a standalone computing system; a region information for the standalone computing system; and a hardware information for the standalone computing system; sending the encrypted request to a respective regional data center of a cloud computing system based on the region information of the first software package; identifying a second software package among a plurality of software packages that are stored in the respective regional data center, wherein the second software package corresponds to the hardware information of the first software package; comparing the version information of the first software package to a version information of the second software package; in response to identifying that the second software package is more recent than the first software package, sending the second software package to the standalone computing system; and deploying the second software package in the standalone computing system.
9. The method of claim 8, further comprising: generating an encrypted response based on the encrypted request; sending the encrypted response to the standalone computing system; and decrypting the encrypted response.
10. The method of claim 8, wherein the first hardware management module of the first software package is different from the second hardware management module of the second software package.
11. The method of claim 8, wherein the first software package is different from the second software package.
12. The method of claim 8, wherein the standalone computing system comprises an automated teller machine (ATM).
13. The method of claim 8, wherein deploying the second software package comprises replacing the first software package with the second software package.
14. The method of claim 8, further comprising: before sending the second software package to the standalone computing system, encrypting the second software package.
15. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to: receive a request from a user; encrypt the request to generate an encrypted request, wherein the encrypted request comprises: a version information for a first software package deployed in a standalone computing system; a region information for the standalone computing system; and a hardware information for the standalone computing system; send the encrypted request to a respective regional data center of a cloud computing system based on the region information of the first software package; identify a second software package among a plurality of software packages that are stored in the respective regional

data center, wherein the second software package corresponds to the hardware information of the first software package; compare the version information of the first software package to a version information of the second software package; in response to identifying that the second software package is more recent than the first software package, send the second software package to the standalone computing system; and deploy the second software package in the standalone computing system.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: generate an encrypted response based on the encrypted request; send the encrypted response to the standalone computing system; and decrypt the encrypted response.

17. The non-transitory computer-readable medium of claim 15, wherein the first hardware management module of the first software package is different from the second hardware management module of the second software package.

18. The non-transitory computer-readable medium of claim 15, wherein the first software package is different from the second software package.

19. The non-transitory computer-readable medium of claim 15, wherein the standalone computing system comprises an automated teller machine (ATM).

20. The non-transitory computer-readable medium of claim 15, wherein deploying the second software package comprises replacing the first software package with the second software package.
